Royal Enfield Tripper BLE Protocol Reverse Engineering Report

Project Goal

Create complete protocol documentation, an open-source SDK, BLE emulator, navigation packet generator and cross-platform client capable of talking to the Royal Enfield Tripper pod without the official application.

Known BLE Information

Service UUID: 01FF0100-BA5E-F4EE-5CA1-EB1E5E4B1CE0.
Characteristic UUID: 01FF0101-BA5E-F4EE-5CA1-EB1E5E4B1CE0.
Notifications are enabled using setCharacteristicNotification() and writeDescriptor() with ENABLE_NOTIFICATION_VALUE.

Core Classes

TripperBleManager: BLE connection, service discovery, notifications, packet queueing, authentication flow and high-level API.
Discovered methods: sendNav(), sendCompass(), sendPin(), sendSetTime(), sendSetTimeNow(), sendPingFirmware(), sendPingWaypoint(), sendShowPin(), sendClose(), sendStopNav(), sendNavIdle(), sendRawPacket(), sendLoadingScreen(), sendLocked().

TripperProtocol: packet generation, parsing, CRC generation and response decoding.

Packet Format

Packets are exactly 20 bytes long. Bytes 0 to 17 contain the payload and bytes 18 and 19 contain the CRC16 checksum. All outgoing packets are passed through enqueuePacket(byte[] data, String label).

CRC

CRC-16/CCITT-FALSE.
Polynomial: 0x1021.
Initial value: 0xFFFF.
RefIn: false.
RefOut: false.
XorOut: 0x0000.
CRC bytes are stored big-endian: packet[18] = high byte and packet[19] = low byte.

Notification Flow

BLE Notification -> handleNotification() -> parseResponse() -> callbacks.
Known response labels: AUTH, OS_VERSION, SERIAL, SESSION, NAV_ACK, TIME_ACK and NACK.

Authentication

Observed responses include AUTH accepted and AUTH rejected. Callbacks exist for onPinAccepted() and onPinRejected(). The currently hypothesized flow is: Connect -> Enable notifications -> Show PIN -> Send PIN -> AUTH -> Ready.

Known Commands

HS_SHOW_PIN = 0x01.
CMD_PING_FW = 0x03.
CMD_PING_WP = 0x30.
The APK references strings including SHOW PIN (0x21 01), PING FW (0x03) and PING WP (0x30).

Navigation

Discovered API: sendNav(byte screen, int dist, byte maneuver, byte heading, byte speedFlags, byte roadType, byte eta). Encoding of the fields remains partially unknown.

Compass

sendCompass(byte direction) exists. Likely mappings correspond to N, NE, E, SE, S, SW, W and NW, but the exact byte values remain unknown.

Firmware

The protocol parses OS_VERSION responses and stores the value in lastTripperFirmware. Firmware version 1.3 has been observed.

Current Reverse Engineering Progress

BLE Discovery: 100%. CRC: 100%. Notifications: 90%. Authentication: 75%. Packet Queue: 100%. Firmware: 90%. Compass: 50%. Time Sync: 50%. Navigation: 35%. Overall SDK completion is estimated at around 60%.

Open Questions

Exact handshake sequence, PIN packet format, time synchronization packet, compass encoding, complete navigation packet structure, complete parser mapping and notification descriptor verification.

End Goal

Produce protocol.md, packets.py, parser.py, ble.py, emulator.py and a complete open-source implementation capable of pairing, authenticating, synchronizing time, displaying navigation and emulating the Royal Enfield Tripper pod.